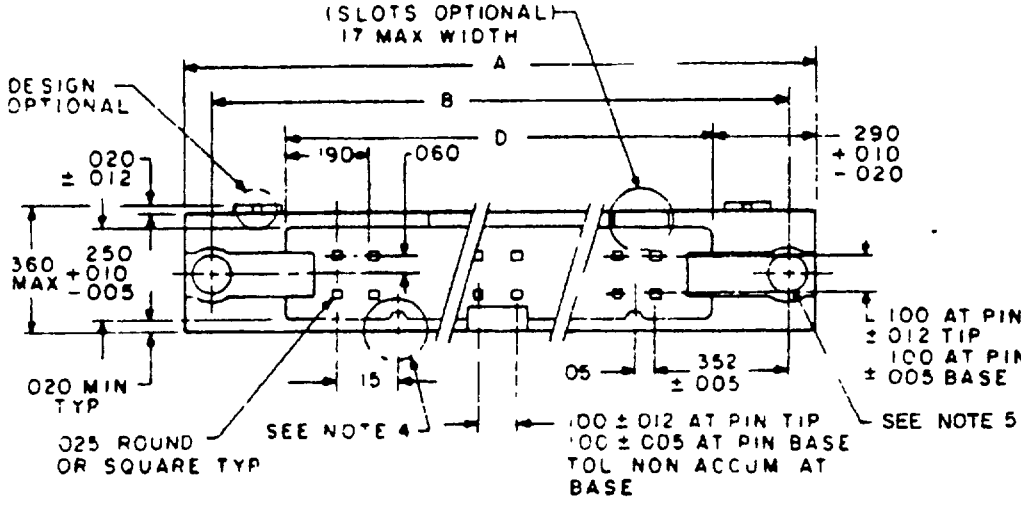
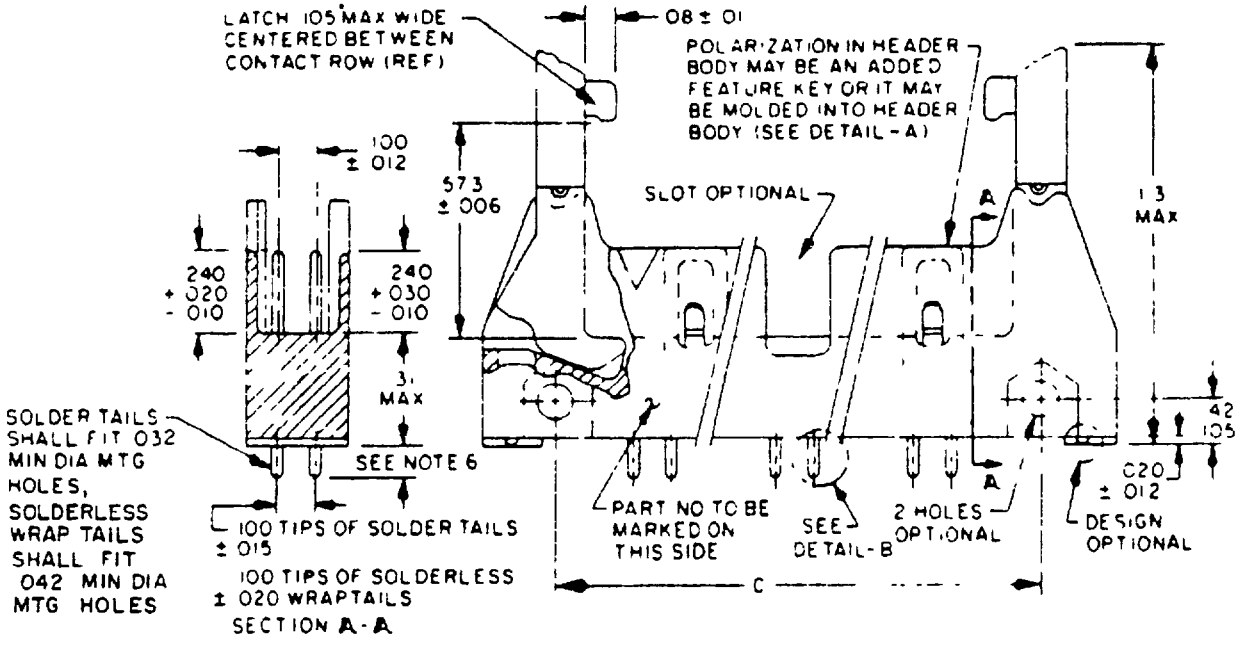
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL, FOUR WALL HEADER, STRAIGHT THRU (.100 SPACING) WITH POLARIZING FEATURE, SOLDERLESS WRAPPOST OR PWB TERMINATION

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

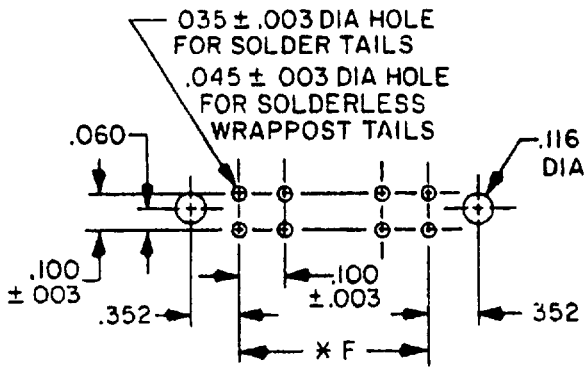
The complete requirements for acquiring the header connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-83503(USAF).





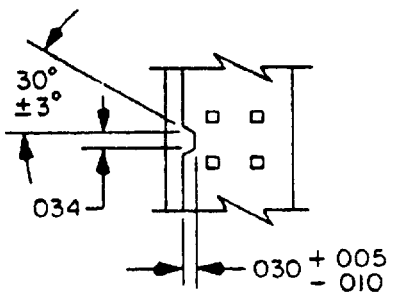
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FIGURE 1. Dimensions and configuration.

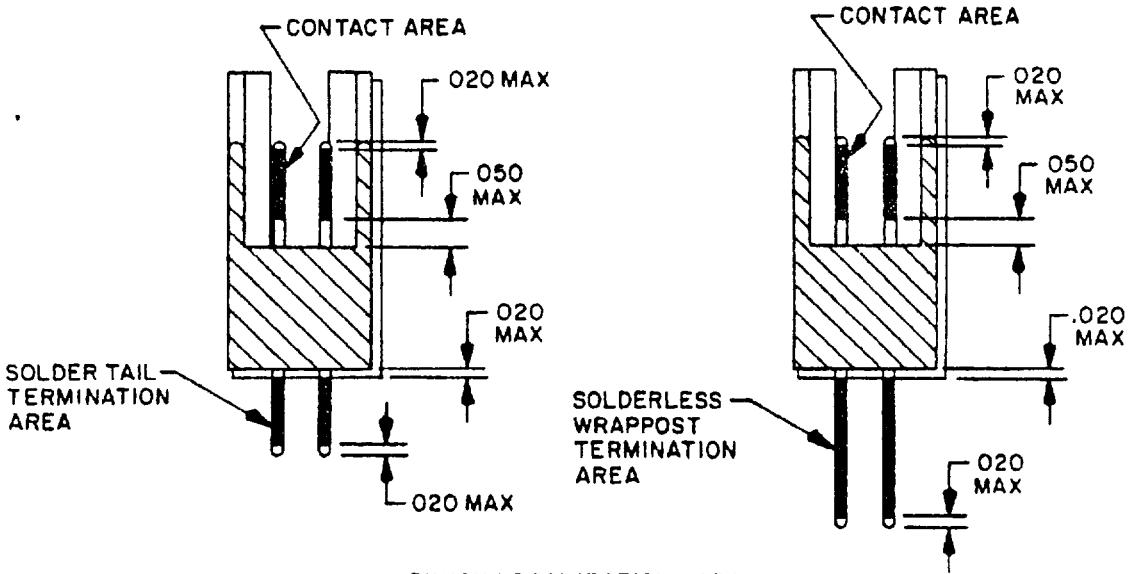


* F = E SPACES X 100
TOL NON ACCUM

RECOMMENDED
MOUNTING HOLE PATTERN



POLARIZATION DETAIL
DETAIL-A



FINISH LOCALIZATION AREA
DETAIL-B

FIGURE 1. Dimensions and configuration - Continued

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Inches	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.030	.76	.089	2.26	.190	4.83
.007	.18	.034	.86	.100	2.54	.210	5.10
.006	.15	.035	.89	.105	2.67	.250	6.35
.010	.25	.045	1.14	.116	2.95	.290	7.37
.01	.3	.05	1.3	.125	3.18	.31	7.9
.012	.30	.050	1.27	.15	3.8	.352	8.94
.015	.38	.060	1.52	.152	3.86	.360	9.14
.020	.51	.062	1.57	.17	4.3	.573	14.55
.025	.63	.08	2.0	.178	4.52	.610	15.49
						1.3	33.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ± 01 (.3 mm) for two place decimals
4. The polarizing feature on this end is non-existent on the 10 position connector and 14 position connector
5. Two vertical mounting studs or two thread cutting screws, No 4-24 type BG are provided.
6. For .062 PWB pin length shall be .089 minimum and .116 maximum, for .125 PWB, pin length shall be .152 minimum and .178 maximum, and for solderless wrappost, pin length shall be .610 ± 015

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number	No. of pins	Dimensions 1/ A	B	C	D +.035 -.000	E	F	Termination, straight contact type
-01	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	062 PWB
-02	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	.125 PWB
-03	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	Wrappost
-04	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	062 PWB
-05	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	.125 PWB
-06	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	Wrappost
-07	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	.062 PWB
-08	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25 15)	7	.700 (17.78)	.125 PWB
-09	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	Wrappost
-10	20	1.76 (44.7)	1 60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	.062 PWB
-11	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	125 PWB
-12	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22,86)	Wrappost
-13	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	.062 PWB
-14	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	.125 PWB
-15	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	Wrappost
-16	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	.062 PWB
-17	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	.125 PWB
-18	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	Wrappost
-19	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	.062 PWB
-20	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	.125 PWB
-21	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	Wrappost
-22	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2 400 (60.96)	.062 PWB
-23	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	.125 PWB
-24	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60 96)	Wrappost
-25	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2.900 (73.66)	.062 PWB
-26	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2 900 (73.66)	.125 PWB
-27	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2.900 (73.66)	Wrappost
-28	64	3.96 (100.6)	3.80 (96 5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	062 PWB
-29	64	3.96 (100.6)	3.80 (96.5)	3 56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	125 PWB
-30	64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	Wrappost

1/ Metric equivalents are in parentheses.

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REQUIREMENTS

Design and construction

Dimensions and configuration: See figure 1 and table I.

Temperature range -55°C to +120°C

Voltage rating 300 V rms.

Current rating. 1 ampere.

Materials:

Contacts Phosphor bronze or brass per CDA260, spring temper, or copper alloy CA-725 per ASTM-B-122.

Contact plating

Contact area Gold.

Solderless wrappost termination area Per MIL-STD-1130

Solder tail termination area. Gold or tin-lead.

Housing: Glass filled polyester per MIL-M-24519, type GPT-15F, GPT-20F or GPT-30F.

Accessories Parts may be constructed of polyester per MIL-M-24519 or other materials specified in MIL-C-83503, provided these parts do not come in contact with current carrying components

Polarization key Shall be permanently attached to header (added feature key shall be bonded to header)

Mating connector See MIL-C-83503/7.

Solderless wrappost contacts Geometry parallelism and tip configuration shall be in accordance with MIL-STD-1130.

Solderless wrappost contact wire accommodation 26 or 28 AWG solid.

Dielectric withstanding voltage (at sea level) 500 V rms, minimum.

(at altitude) 200 V rms, minimum

Cable retention Not applicable.

Contact retention Not applicable.

Connector-cable flexing Not applicable.

Contact resistance: 50 milliohms, maximum.

Latch retention The latch shall retain the mating connector when the cable retention force is applied to the mating connector cable. The latch shall also function as an ejector to eject mating connector

Marking

Part number M83503/25-(dash number from table I).

Contact identification Molded triangle "▲" or notch to show number 1 contact, or raised numbers to show position number "1".

NOTE. Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodian
Air Force - 85

Review activities
Air Force - 11, 17, 99
DLA - ES

Agent
DLA - ES

Preparing activity
Air Force - 85

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